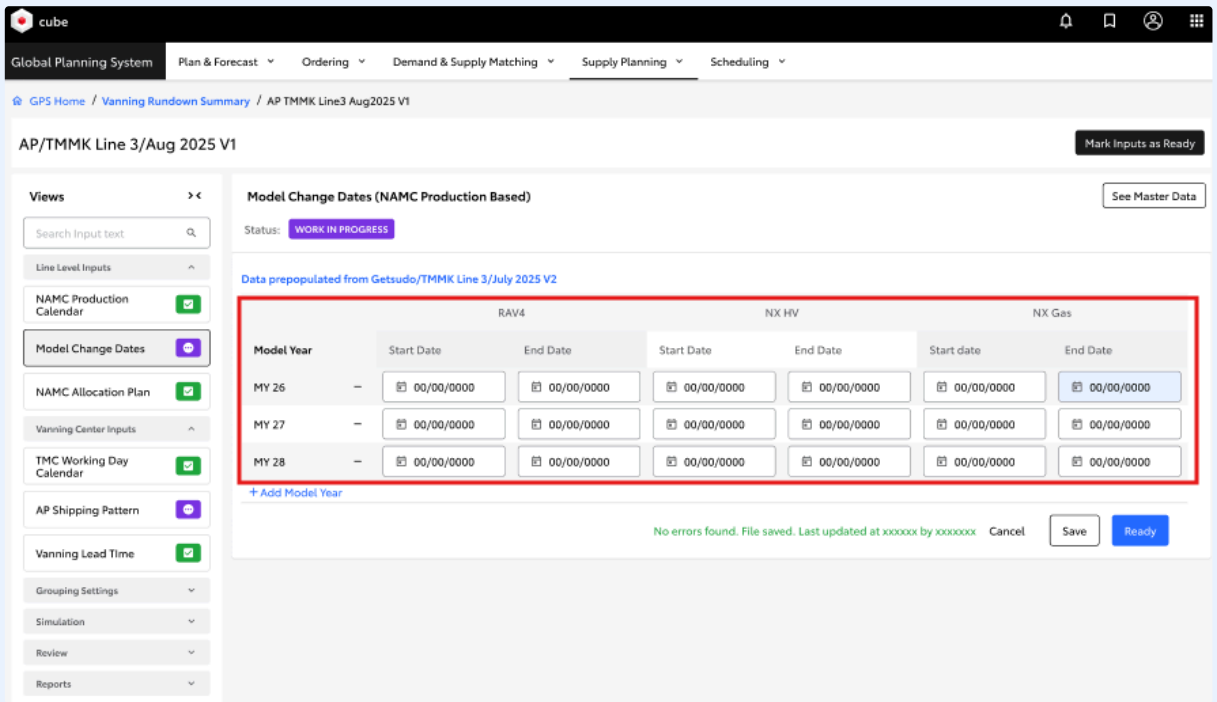
SPPL-192: Model Change Dates Get API technical workflow

Jira Ticket: SPPL-192: BE | Create technical workflow for Model Change Dates Get & Post API on parameter inputs page including DB details ACCEPTED

This page provides comprehensive information about the GET API for Model Change Dates on the Input params page, to fetch model change dates data along with all the steps status data.



Supply Planning GET Model Change Dates API Details:

1. **Hosted URL**: https://api.spln.cube<env>.toyota.com/<env>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/model-change-dates
3. **Path Param**: N/A
4. **Query Param**: ?scenarioId=uniqueScenarioId&userEmail=userEmail
5. **Method:** GET
6. **Authorization**: Bearer token (**required**)
7. **Lambda Name**: spln-<env>-view-model-change-dates-v1-1
8. **API Request Body**: N/A
9. **API Response:**
   - **Success response with status 200:**

```
{
  "scenarioId": "uniqueScenarioId",
```

```
  "data": [
    {
      "subSeries": "RAV4",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "2025-01-01",
          "endProdDate": "2025-12-30"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "2026-01-01",
          "endProdDate": "2026-12-30"
        }
      ]
    },
    {
      "subSeries": "NX HV",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "2025-01-01",
          "endProdDate": "2025-12-30"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "2026-01-01",
          "endProdDate": "2026-12-30"
        }
      ]
    }
  ],
  "scenarioSteps": {
    "Line Level Input": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```


- **Error Response:**
  - **any internal server error occurred with status 500:**


```
{
  "errorMessage": "Internal Server Error"
}
```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
{
  "message": "Unauthorized"
}
```

- **If request is in-valid with status 400:**

```
{
  "errorMessage": [<"ValidationError: validation error message”>]
}
```

- **List of possible validation error message:**
  1. "ValidationError: scenarioId is required and must be a uuid."
  2. "ValidationError: Scenario doesn't exist."
  3. “ValidationError: userEmail is required and must be a string.”
  4. “ValidationError: Invalid userEmail.”

10. **Table:**
    a. **scenarios**
    b. **model_change_date**
    c. **scenario_step_status**

SP Fetch Model Change Dates API Workflow:

- **Technical workflow:**
  - Description:
    i. The Supply Planning user logs in to the application.
    ii. After the user provides creates a scenario, opens it and clicks Model Change Dates, a request is sent to the SP Get Model Change Dates endpoint: ***sp/model-change-dates*** with input param - scenarioId & userEmail.
    iii. After successful authentication, the **spln-view-model-change-dates-v1-1** lambda is triggered by the SP view scenario Model Change Dates data endpoint: ***sp/*model-change-dates**. If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401***) will be returned and sent to client.
    iv. After user authentication has been confirmed, the **spln-view-model-change-dates-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response**: Section if request is in-valid with status 400 and List of possible validation error message***). Below query is used to check if provided scenario is valid:

```
select * from supply_planning.scenarios where scenario_id = :scenarioId and
is_active = true;
```

v. After the request payload is successfully validated, the **spln-view-model-change-dates-v1-1** lambda execute below query to get scenario Model Change Dates data from the **model_change_date** table.

```
select * from supply_planning.model_change_date where scenario_id = :scenarioId;
```

vi. Below query is used to fetch view scenario details (if status data doesn't exist for a scenario, then insert the details): ref(SPLN-162: View Scenario Menu technical workflow document)

```
select * from supply_planning.scenario_step_status where scenario_id =
:scenarioId;
```

vii. If Model Change Dates data exists, format and return the below response to the client with a status code of 200:

```
{
  "scenarioId": "uniqueScenarioId",
  "data": [
    {
      "subSeries": "RAV4",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "2025-01-01",
          "endProdDate": "2025-12-30"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "2026-01-01",
          "endProdDate": "2026-12-30"
        }
      ]
    },
    {
      "subSeries": "NX HV",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "2025-01-01",
          "endProdDate": "2025-12-30"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "2026-01-01",
          "endProdDate": "2026-12-30"
        }
      ]
    }
  ],
  "scenarioSteps": {
    "Line Level Input": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
```

```
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```

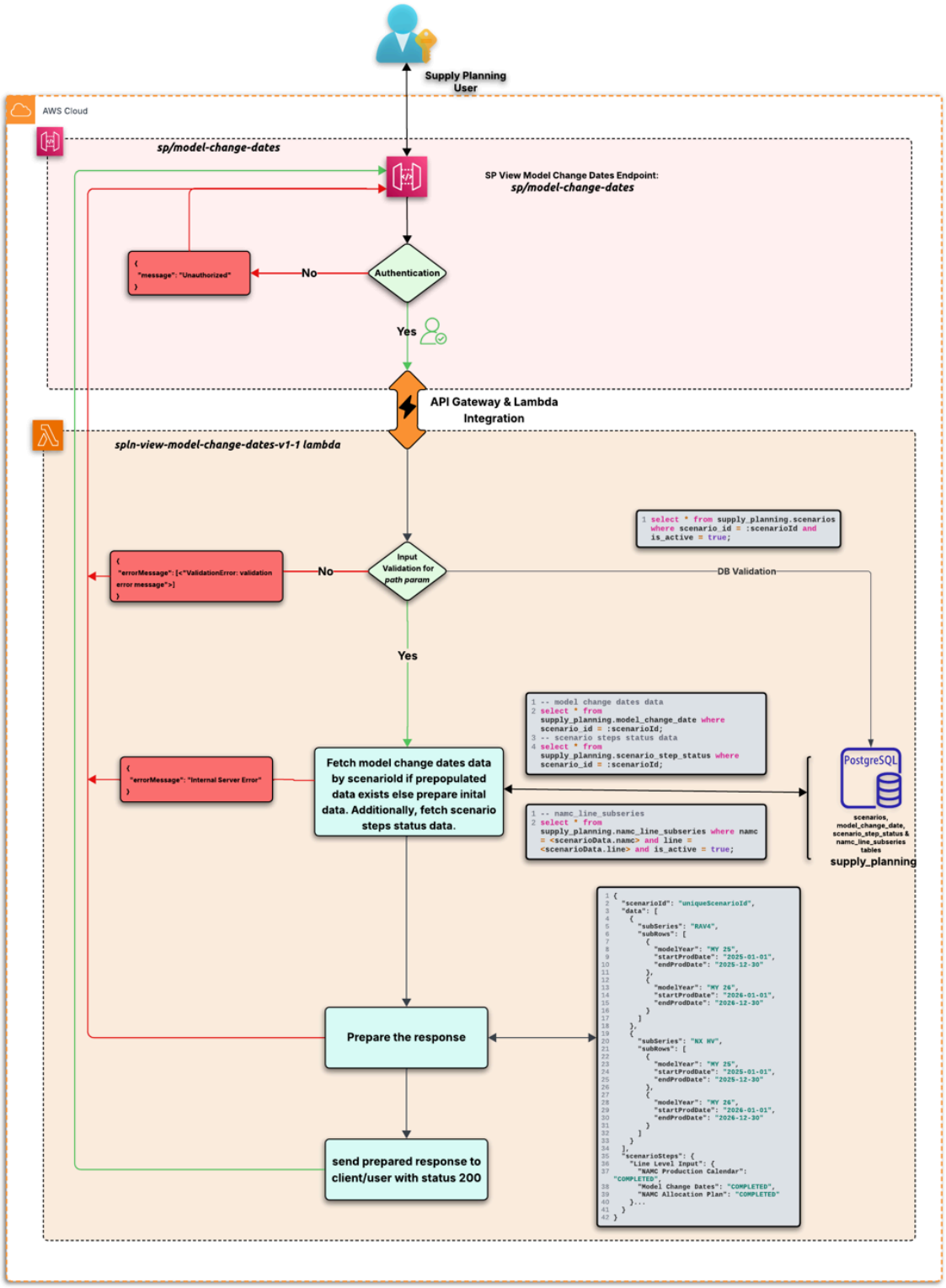
viii. If Model Change Dates data doesn't exists, create initial data based on scenario timeframe (use query (iv) response for timeframe) with below format and return the response to the client with a status code of 200. Use below query to fetch subseries associated with the scenario:

```
select * from supply_planning.namc_line_subseries where namc =
<scenarioData.namc> and line = <scenarioData.line> and is_active = true;
```

```
{
  "scenarioId": "uniqueScenarioId",
  "data": [
    {
      "subSeries": "RAV4",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "0000-00-00",
          "endProdDate": "0000-00-00"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "0000-00-00",
          "endProdDate": "0000-00-00"
        }
      ]
    },
    {
      "subSeries": "NX HV",
      "subRows": [
        {
          "modelYear": "MY 25",
          "startProdDate": "0000-00-00",
          "endProdDate": "0000-00-00"
        },
        {
          "modelYear": "MY 26",
          "startProdDate": "0000-00-00",
          "endProdDate": "0000-00-00"
        }
      ]
    }
  ],
  "scenarioSteps": {
    "Line Level Input": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
```

```
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```

ix. If there is an error while updating the data, an “Internal Server Error” with a status code of 500(*refer: error response-**any internal server error occurred with status 500***) will be returned.



SP Model Change Dates Get API Techncial workflow

Queries: